Recommender Systems
lecture 4: content-based and hybrid systems

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Recommender Systems taxonomy

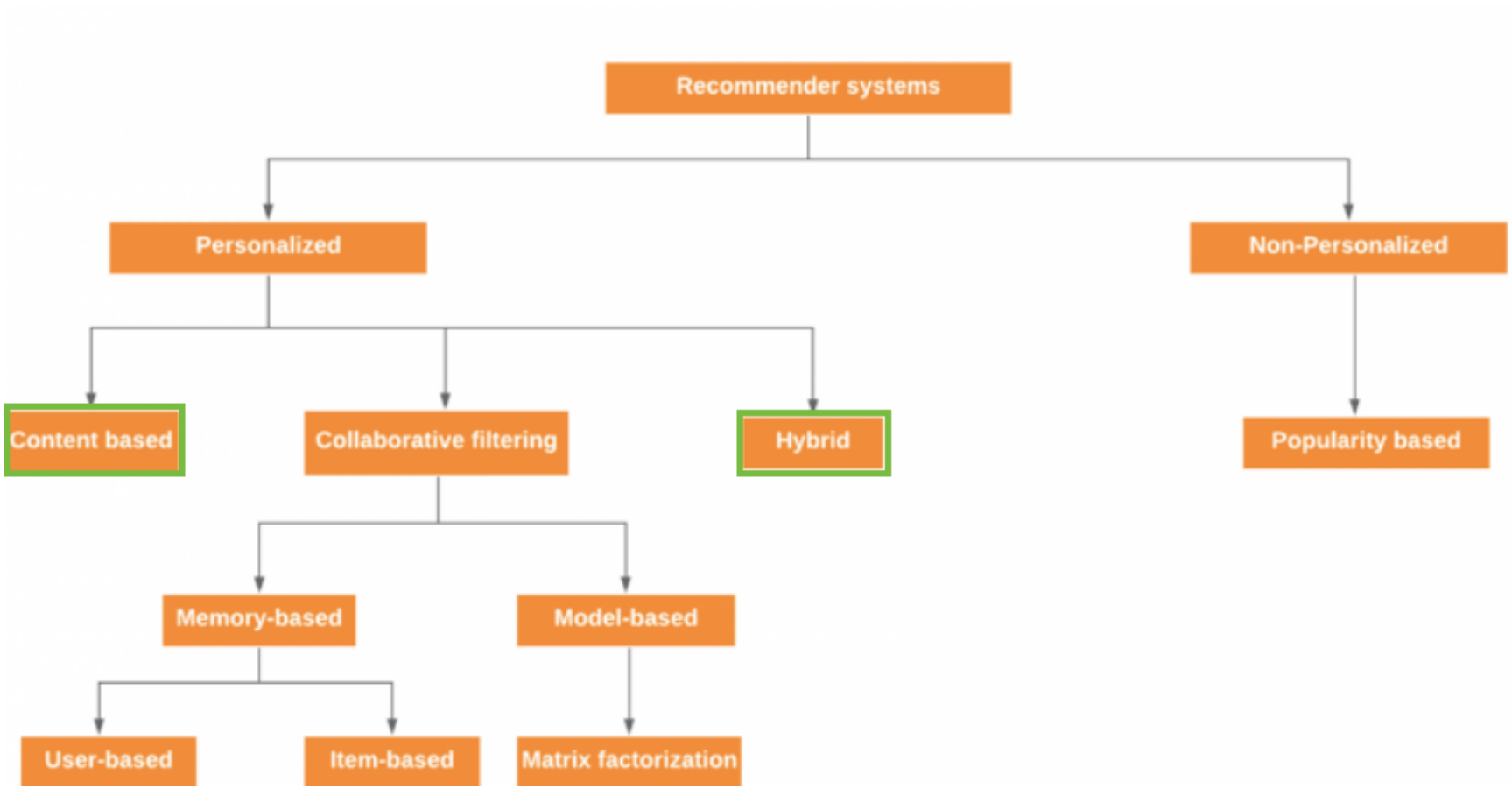


image credit: https://thingsolver.com/introduction-to-recommender-systems

Recommendation techniques

1. Collaborative recommender systems produce user specific recommendations of items based on patterns of ratings or usage without need for exogenous information about either items or users.
2. **A content-based recommender** learns to recommend items that are similar to the ones that the user liked in the past. The similarity of items is calculated based on the features associated with the compared items.
3. Demographic recommender system recommends items based on the demographic profile of the user.
4. Community-based recommender system recommends items based on the preferences of the users friends.
5. **Hybrid recommender systems** are based on the combination of the above mentioned techniques. A hybrid system combining techniques A and B tries to use the advantages of A to fix the disadvantages of B.

F. Ricci, L. Rokach, B. Shapira. (2011). Recommender Systems Handbook.

From matrix factorization to factorization machine

matrix factorization

$$\hat{r}_{ui} =< p_u, q_i >= \sum_{i=1}^{k} p_i q_i$$

factorization machine

$$\hat{r}(\mathbf{x}) = w_0 + \sum_{i=1}^{n} w_i x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} w_{ij} x_i x_j,$$

where $w_0 \in R$, $w \in R^n$, $V \in R^{n \times k}$

Factorization machines

- **Idea (FM):** model all interactions between features using factorized parameters. This allows the model to estimate interactions even in highly sparse data where certain combinations never occurred.
- **Idea (FFM):** An extension where each feature has multiple latent vectors (one for each field).

	Feature vector **x**																					Target y	
	User				Movie					Other Movies rated					Time	Last Movie rated							
	A	B	C	...	TI	NH	SW	ST	...	TI	NH	SW	ST	...		TI	NH	SW	ST	...			
$\mathbf{x}^{(1)}$	1	0	0	...	1	0	0	0	...	0.3	0.3	0.3	0	...	13	0	0	0	0	...		5	$y^{(1)}$
$\mathbf{x}^{(2)}$	1	0	0	...	0	1	0	0	...	0.3	0.3	0.3	0	...	14	1	0	0	0	...		3	$y^{(2)}$
$\mathbf{x}^{(3)}$	1	0	0	...	0	0	1	0	...	0.3	0.3	0.3	0	...	16	0	1	0	0	...		1	$y^{(2)}$
$\mathbf{x}^{(4)}$	0	1	0	...	0	0	1	0	...	0	0	0.5	0.5	...	5	0	0	0	0	...		4	$y^{(3)}$
$\mathbf{x}^{(5)}$	0	1	0	...	0	0	0	1	...	0	0	0.5	0.5	...	8	0	0	1	0	...		5	$y^{(4)}$
$\mathbf{x}^{(6)}$	0	0	1	...	1	0	0	0	...	0.5	0	0.5	0	...	9	0	0	0	0	...		1	$y^{(5)}$
$\mathbf{x}^{(7)}$	0	0	1	...	0	0	1	0	...	0.5	0	0.5	0	...	12	1	0	0	0	...		5	$y^{(6)}$

LightFM

The latent representation of user u is given by the sum of its features' latent vectors:

$$\boldsymbol{q}_u = \sum_{j \in f_u} \boldsymbol{e}_j^U$$

The same holds for item i:

$$\boldsymbol{p}_i = \sum_{j \in f_i} \boldsymbol{e}_j^I$$

The bias term for user u is given by the sum of the features' biases:

$$b_u = \sum_{j \in f_u} b_j^U$$

The same holds for item i:

$$b_i = \sum_{j \in f_i} b_j^I$$

The model's prediction for user u and item i is then given by the dot product of user and item representations, adjusted by user and item feature biases:

$$\widehat{r}_{ui} = f\left(\boldsymbol{q}_u \cdot \boldsymbol{p}_i + b_u + b_i\right) \tag{1}$$

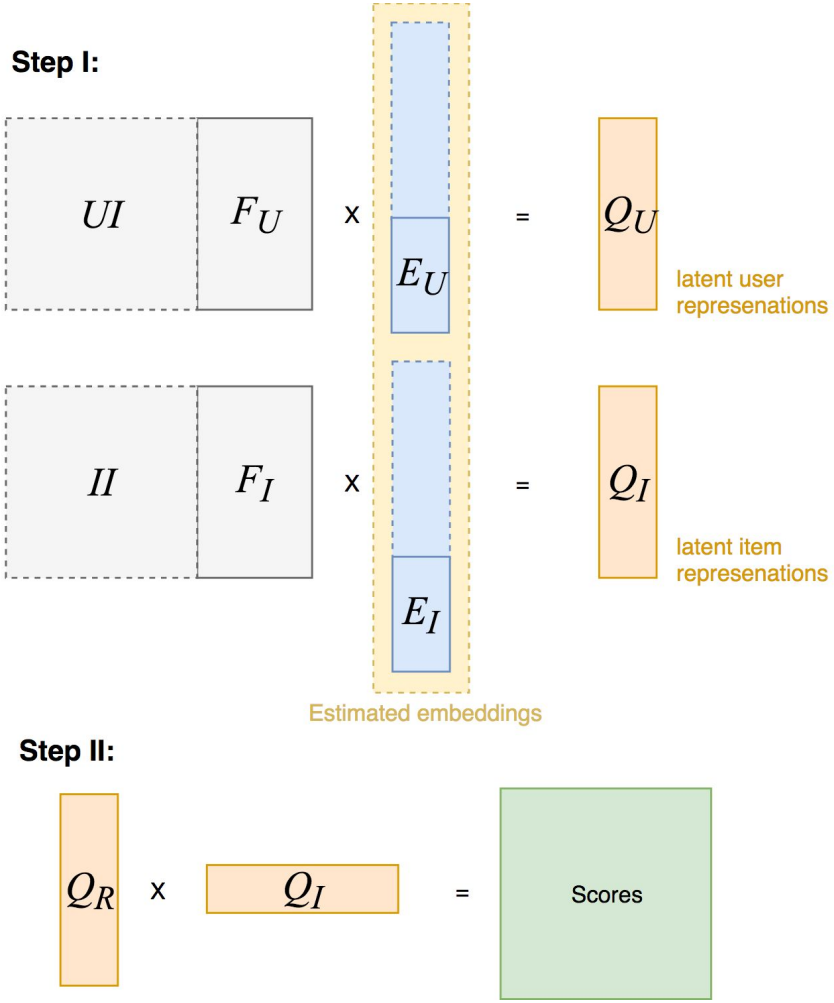
There is a number of functions suitable for $f(\cdot)$. An identity function would work well for predicting ratings; in this paper, I am interested in predicting binary data, and so after Rendle *et al.* [16] I choose the sigmoid function

$$f(x) = \frac{1}{1 + \exp(-x)}.$$

The optimisation objective for the model consists in maximising the likelihood of the data conditional on the parameters. The likelihood is given by

$$L\left(\boldsymbol{e}^U, \boldsymbol{e}^I, \boldsymbol{b}^U, \boldsymbol{b}^I\right) = \prod_{(u,i) \in S^+} \widehat{r}_{ui} \times \prod_{(u,i) \in S^-} (1 - \widehat{r}_{ui}) \tag{2}$$

LightFM in matrix form



$$r_{ui} = \langle q_u, p_i \rangle + b_u + b_i$$

$$q_u = \sum_{j \in f_u} e_j^U$$

$$e_j^U = w_j^U e_j$$

$$b_u = \sum_{j \in f_u} b_j^U$$

$$b_j^U = w_j^U b_j$$

image credit: https://datarevenue.webflow.io/en-blog/building-a-production-ready-recommendation-system

BPR (Bayesian Personalized Ranking)

- **Data:** consider triplets $D_S := \{(u, i, j) \mid i \in I_u^+ \wedge j \in I \setminus I_u^+\}$ of user u, positive item i and negative/unkwown item j.
- **Objective:** We want to score the positive item higher than the negative one.

Optimization Target

$$\mathcal{L} = \sum_{(u,i,j) \in D_S} \ln \sigma \left(\hat{x}_{ui}(\theta) - \hat{x}_{uj}(\theta) \right) - \lambda_\theta \|\theta\|^2 \to \max$$

WARP Loss

Idea: sampling until violation

Instead of picking one random negative item (like BPR), we sample negative items j until we find a **violator**:

$$f_u(j) > f_u(i) - 1,$$

where $f_u(i)$ — model score, e.g. $< q_u, p_i >$

If we found a violator on the M-th attempt, the rank of item i can be estimated as:

$$\text{rank}_i(f_u) \approx \left\lfloor \frac{Y-1}{N} \right\rfloor$$

Final Loss Step:

$$L\left(\text{rank}_i(f_u)\right) |1 - f_u(i) + f_u(j))|_+,$$

where $L(k)$ transforms rank into loss (e.g., $\sum_{i=1}^{k} \frac{1}{i}$).

WARP loss computation

Deriving differential rank approximation

$$
\begin{aligned}
L(\mathrm{rank}_i(f_u)) &= L(\mathrm{rank}_i(f_u))\frac{\mathrm{rank}_i(f_u)}{\mathrm{rank}_i(f_u)} \\
&= L(\mathrm{rank}_i(f_u))\frac{\sum_{j \notin I_u^+} I(f_u(j) \geq f_u(i))}{\mathrm{rank}_i(f_u)} \\
&= \sum_{j \notin I_u^+} L(\mathrm{rank}_i(f_u))\frac{I(f_u(j) \geq f_u(i))}{\mathrm{rank}_i(f_u)} \\
&= \sum_{j \notin I_u^+} L(\mathrm{rank}_i(f_u))\frac{|1 - f_u(i) + f_u(j)|_+}{\mathrm{rank}_i(f_u)}
\end{aligned}
$$

Algorithm 1 Online WARP Loss Optimization

Input: labeled data (x_i, y_i), $y_i \in \{1, \ldots, Y\}$.
repeat
  Pick a random labeled example (x_i, y_i)
  Let $f_{y_i}(x_i) = \Phi_W(y_i)^\top \Phi_I(x_i)$
  Set $N = 0$.
  repeat
    Pick a random annotation $\bar{y} \in \{1, \ldots, Y\} \setminus y_i$.
    Let $f_{\bar{y}}(x_i) = \Phi_W(\bar{y})^\top \Phi_I(x_i)$
    $N = N + 1$.
  until $f_{\bar{y}}(x_i) > f_{y_i}(x_i) - 1$ or $N \geq Y - 1$
  if $f_{\bar{y}}(x_i) > f_{y_i}(x_i) - 1$ **then**
    Make a gradient step to minimize:
      $L(\lfloor \frac{Y-1}{N} \rfloor)|1 - f_y(x_i) + f_{\bar{y}}(x_i)|_+$
    Project weights to enforce constraints (2)-(3).
  end if
until validation error does not improve.

Literature

1. *F. Ricci, L. Rokach, B. Shapira.* (2011). Recommender Systems Handbook.
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4. *Steffen Rendle.* (2010). Factorization Machines.
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6. https://github.com/lyst/lightfm
7. *A. Milogradskii et al.* (2024). Revisiting BPR: A Replicability Study of a Common Recommender System Baseline